SEC 2026 · MINIMALIST EDGE AI CHALLENGE

Prune deeper. Recover smarter.

Structured pruning, teacher-guided recovery, static INT8 —
on a MobileNetV2 that was already distilled.

8.94×
faster on the edge CPU

5.58×
fewer parameters

98.0%
of accuracy kept

VALIANT JOY | MobileNetV2 / EuroSAT | Software Optimization Track

THE PIPELINE

Four steps, one of them unusual

01
Provided student
MobileNetV2 FP32
2.24 M params

→

02
Structured pruning
60% of channels removed, not masked

→

03
Teacher recovery
BN recalibration + 8 KD epochs from the provided ResNet-50

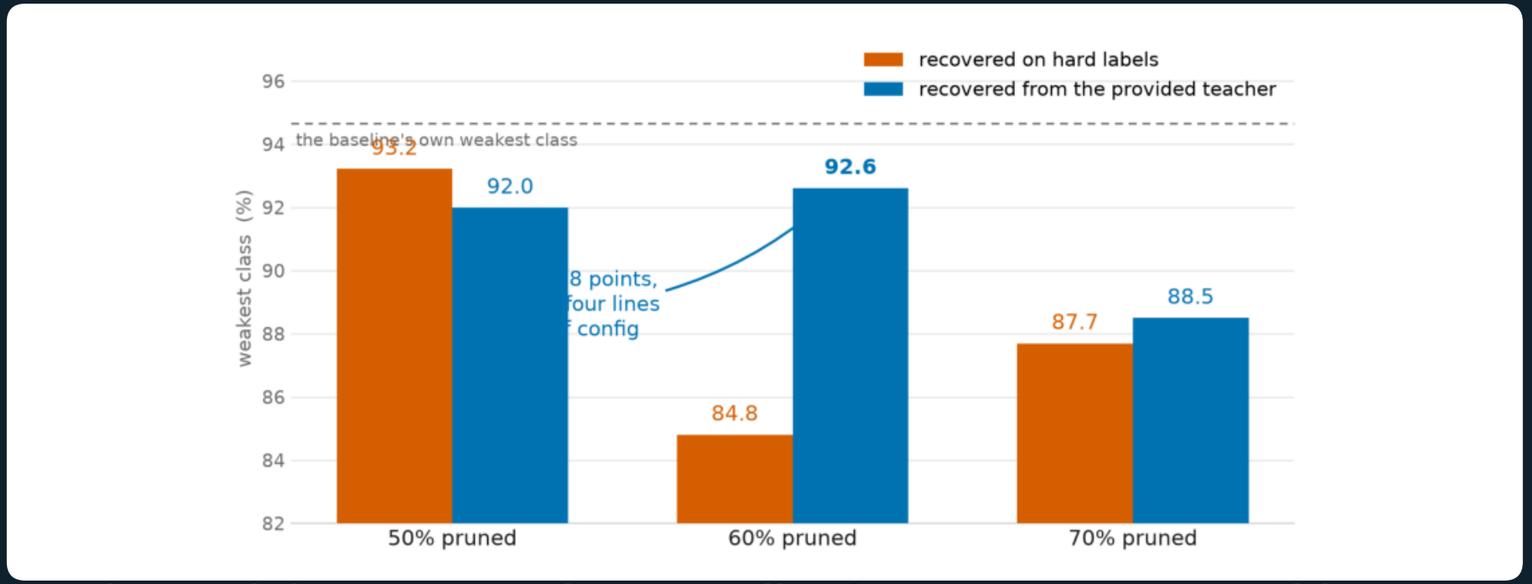
→

04
Static INT8
QDQ, per-channel, percentile calibration

Exported artifact: ONNX opset 13 · static QDQ · input pinned to 1×3×224×224 · 0.60 MB

THE FINDING

The collapse at 60% is a recovery limit, not a capacity limit



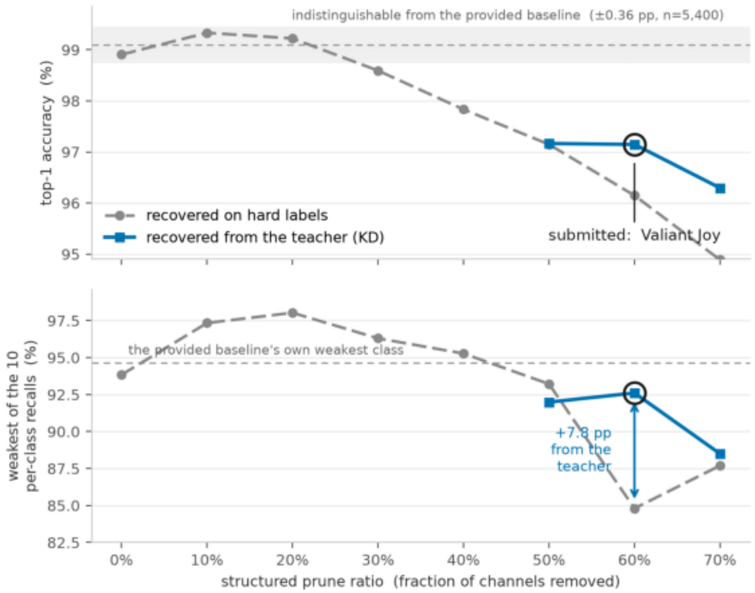
+7.8

points of the weakest class at 60% — for four lines of config. And nothing at all at 50%.

Weakest of the ten per-class recalls. Same cut, same architecture, same quantization. The benefit peaks in a window: too shallow and hard labels are already enough; too deep and there is no capacity left to teach.

HOW WE CHOSE THE POINT

We swept it. The mean could not pick a stopping point



Top panel

every point from 0% to 20% is inside the ±0.36-point measurement band. The bump at 10% is noise.

Bottom panel

the weakest class is the statistic that moves, and the one that located the cliff between 50% and 60%.

Both recoveries

plotted together — the blue line is what let us ship the deeper point.

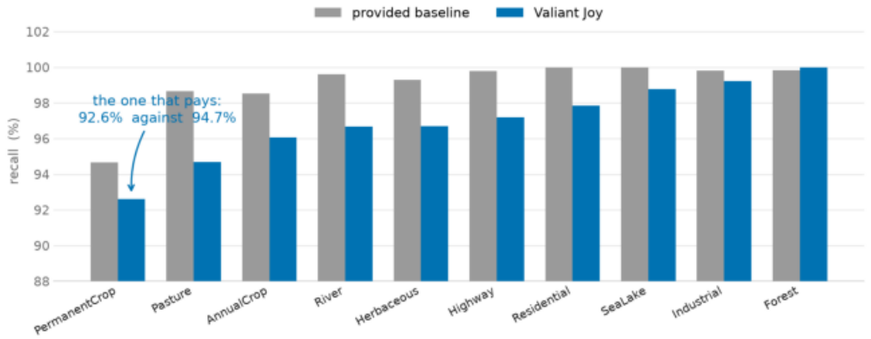
ACCURACY AND COMPRESSION

82% of the parameters gone, 98% of the accuracy kept

98.04%
of baseline accuracy
99.09% → 97.15% top-1

5.58×
fewer parameters
2.24 M → 0.40 M

14.5×
smaller on disk
8.77 MB → 0.60 MB



What it costs

The weakest of the ten classes falls to 92.6%, below the baseline's own 94.7%. One class carries most of the loss, and we report it rather than averaging it away.

RESOURCE AND ENERGY

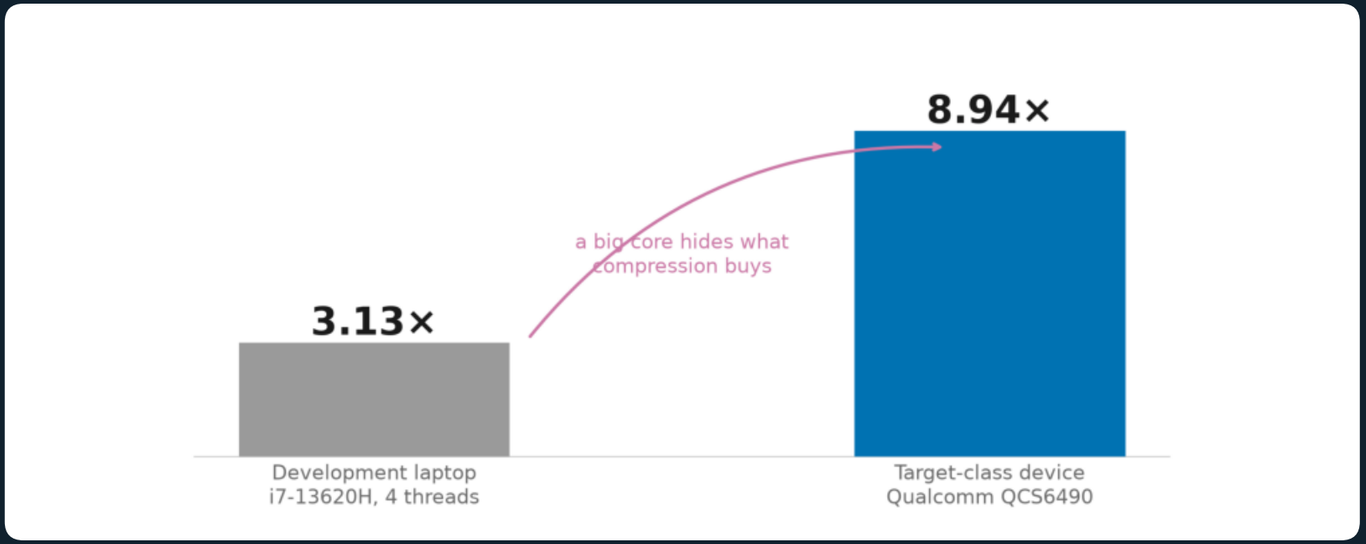
Less compute, less memory, less energy

Metric	Baseline	Valiant Joy	Change
MACs	0.326 G	0.080 G	**4.10×**
CPU latency, 4 threads	3.50 ms	1.12 ms	**3.13×**
Energy per inference	0.218 J	0.071 J	**3.07×**
Peak RAM, raw process RSS	78.9 MB	62.1 MB	**1.27×**

Same machine, same script, same thread count for baseline and final. Energy is CodeCarbon in TDP mode — so we checked it against physics: implied power 62.2 W for the baseline and 63.4 W for ours, and a 3.07× energy ratio against a 3.13× latency ratio. Peak RAM moves least, because 56 MB of the process is the Python and ONNX Runtime floor, not the model.

HARDWARE-AWARE EVALUATION

Our laptop said 3.1×. The edge CPU said 8.9×.



LATENCY THERE
27.97 ms → 3.13 ms

PEAK MEMORY THERE
32.33 MB → 19.45 MB (1.66×)

PROFILED ON
Dragonwing RB3 Gen 2 QCS6490, ONNX on CPU

Same two ONNX files, profiled on both. A large out-of-order core with a deep cache absorbs the cost of a fat model and hides most of what compression buys. Choose a prune ratio on your laptop and you are reading a gradient three times too shallow.

ENGINEERING DECISIONS

What we learned, including what failed

01

Measure the weakest class

The mean loses one point where the worst class loses eight. The cliff is invisible in the average.

02

Spend on recovery, not the cut

Teacher-guided recovery moved the deepest usable prune a full step down the ladder.

03

Profile on the target

The same two files read 3.13× on our laptop and 8.94× on the edge CPU.

04

What did not work

Per-layer allocation: better at matched size, but it stalls at 46.6% — cheap parameters are a bounded pool.

98.04% accuracy retained · 5.58× fewer parameters · 14.5× smaller · 8.94× faster

Every figure traces to one CSV from one run. Both ladders, the failed experiments and the conservative alternative ship with the code.

THE MEASUREMENTS

Everything we measured, in full

Measurement	Baseline	Valiant Joy	Ratio
Top-1 accuracy	99.09 %	97.15 %	**98.04 % retained**
Weakest of 10 class recalls	94.66 %	92.61 %	−2.05 pp
Parameters	2,236,682	400,954	**5.58×**
ONNX file size	8.51 MB	0.60 MB	**14.07×**
MACs	326.2 M	79.5 M	**4.10×**
Latency, 1 thread	9.804 ms	2.679 ms	**3.66×**
Latency, 4 threads	3.501 ms	1.119 ms	**3.13×**
CPU time per inference	12.308 ms	4.646 ms	**2.65×**
Peak RAM, raw process RSS	78.85 MB	62.14 MB	**1.27×**
Energy per inference	0.2177 J	0.0709 J	**3.07×**
Implied power (energy ÷ latency)	62.2 W	63.4 W	1.02× — the check
AI Hub latency, QCS6490	27.969 ms	3.127 ms	**8.94×**
AI Hub peak memory	32.33 MB	19.45 MB	**1.66×**

THE SCORECARD

Every number the rubric asks for

Criterion	Pts	What is measured	Valiant Joy
Model Accuracy & Reliability	25	retained vs. baseline, all 10 classes	**98.04% retained worst class 92.6%**
Model Compression	20	parameter count and file size	**5.58× parameters 14.51× file**
Resource Efficiency	20	peak RAM and latency, cross-checked on AI Hub	**8.94× faster on QCS6490 1.66× less peak memory**
Energy Parsimony	20	Joules per inference, ratio to own baseline	**3.07× less 0.2177 J → 0.0709 J**
Documentation & Reproducibility	10	rebuild from the provided checkpoint	**four commands, every figure one CSV**
Engineering Judgment	5	quality of the tradeoff analysis	**four documented negative results**